On a Simple Relationship Between Order Imbalance, Skew and Width in Over-The-Counter Trading

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Abstract

A market maker fills sealed-bid enquiries that arrive with imbalanced intent: sellers more often than buyers, or the reverse. When the best competing quote is exponentially distributed around a commonly discerned fair price, the steady state problem facing imbalanced flow compresses exactly onto the balanced one: imbalance is absorbed by a translation of the skew together with a uniform slowing of the inventory clock — equivalently, multiplication of the carrying cost by $1/(2\sqrt{q(1-q)})$. The correction involves no parameter beyond the observable market width, and the exponential assumption is needed only on the strikes the transformation traverses. Among the consequences: a dealer with zero inventory and symmetric carrying costs still skews; skew responds to imbalance at first order and width only at second; and constant-width linear-skew quoting reappears as the small-skew case of balanced flow with quadratic cost.

Keywords: market making, order imbalance, skew, bid–ask spread, request for quote, inventory cost.
MSC codes: 91G15, 93E20.

1 Introduction

The facilitation of trade by intermediaries often takes the form of a sequence of sealed-bid auctions. A customer wishing to sell an item will typically obtain several bids from dealers and select the best price; a customer wishing to buy will perform a similar search and transact with whomever offers the lowest. The market maker takes on price risk and carrying cost, and must respond to each enquiry with that in mind. She is said to mark up her offer, or mark down her bid, relative to a fair market price.

We have in mind fungible goods for which this description is apt (dealer markets in physical commodities, corporate and municipal bonds, blocks of securities, or for that matter rare baseball cards) and one empirical regularity of such markets that the standard theory does

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not deliver. Dealers shade their quotes with the direction of the flow. A dealer who expects the next call to be a seller quotes differently from one who expects a buyer, and does so even when her book is flat. Practitioners regard this as too obvious to remark upon. The optimal market making literature has meanwhile concentrated on the other lever: the benchmark policies descending from Avellaneda and Stoikov [4] hold width constant and skew linearly in inventory, with buy and sell arrivals assumed symmetric.

The purpose of this paper is to show that flow-shading is not a heuristic layered on top of inventory management but an exact feature of optimal behavior, and that it follows from a symmetry. In the steady state, the market making problem facing imbalanced flow compresses onto the perfectly balanced problem: the imbalance is absorbed, without approximation, by a translation of the skew together with a uniform slowing of the inventory clock, which multiplies positive carrying costs by an explicit constant. The adjustments involve nothing but the imbalance itself and the market width.

The solution being adjusted is not simple. The indifference cost that characterizes optimal behavior must in general be computed numerically, and its shape depends on the entire cost of carry function. The imbalance correction nevertheless rides on top of any such solution unchanged, so a single balanced solver covers every level of imbalance, one solve per multiplier. Nor is the exponential assumption load-bearing in its global form: only the hazard at the quotes actually made enters, a point Remark 6 makes precise.

We note three consequences. A market maker with zero inventory and symmetric carrying costs should still skew, by half the market width times the log-odds of the arrival direction, which is approximately twice the market width times the imbalance when the imbalance is mild. The width response is second order in the imbalance while the skew response is first order, which we believe is why practitioners skew before they widen. And the constant width, linear skew heuristic reappears in the special case of balanced flow and quadratic holding cost, which locates what that benchmark omits: the flow term. The skew magnitudes involve no free parameter beyond the observable width; the width response involves in addition the balanced policy's cost-scale sensitivity, for which we give a closed small-skew approximation. Both are falsifiable in principle; we make no empirical claims here.

1.1 The constant-width, linear-skew benchmark

To fix terminology, a dealer's *width* is the difference between where she is prepared to bid and where she is prepared to offer, and her *skew* is her estimate of fair value minus the average of the two, so that positive skew means a downward lean. The Constant Width Linear Skew heuristic maintains a fixed width and sets skew proportional to signed inventory, with a sign that leans the dealer towards a neutral position.

Motivations for CWLS range from its prima facie reasonableness to formal justification as an approximate solution of a stochastic control problem [4, 5]. These justifications require some approximation, however, and they assume symmetric flow throughout. From a practical perspective CWLS is not a straw man, because it is simple, and because it exactly solves the interior equations in a particular compatible family. Section 4 makes the second point precise, and the symmetry of Section 3 supplies the benchmark's missing column: what to do about imbalance.

1.2 Literature

The dealer-inventory line begins with Garman [1], whose dealer faces genuinely asymmetric Poisson buy and sell rates, the oldest imbalanced-flow model, but sets a single static price pair, the content being ruin rather than policy. Amihud and Mendelson [2] are the closest classical antecedent in spirit: imbalanced arrivals move a preferred inventory position and monotone quotes, though the results are structural rather than closed form. Ho and Stoll [3] quote from value function differences, and the *slope* half of the slope-and-convexity characterization of Section 2.2 is implicit there and in everything since; the *convexity* half states that the discretionary component of a dealer's width reveals the second difference of her inventory cost exactly, so that her inventory cost can be read off her quoting. We have not found the latter stated elsewhere.

Avellaneda and Stoikov [4] anchor the modern literature. Their constant width, linear skew policy is reached by dropping terms, and the dropped terms are precisely the ones needed to establish any rule regarding width. Constant width is an artifact of the approximation, not a conclusion of the model, and the line consequently has nothing to say about how width should respond to inventory or to flow.

The exact treatment of Guéant, Lehalle and Fernandez-Tapia [5] takes arrival intensities to be symmetric; the one directional asymmetry handled in closed form in that line is drift in the mid-price, which shifts quotes but neither widens them nor taxes carry. Guéant [6, 7] and Bergault et al. [13, 14] accommodate asymmetric intensities in generality but handle them numerically. We have found no equivalence statement, no log-odds shift, and no cost multiplier in that literature.

Cartea, Jaimungal and coauthors [8, 9, 10] treat order-flow imbalance as a predictive signal, a different object from the stationary structural imbalance considered here. Bergault and Guéant [18] observe numerically that flow-aware market makers skew their quotes even in the absence of inventory; Corollary 2 is the closed form of that observation.

The request-for-quote literature [11, 12, 15] arrived some years after this work was completed. Fermanian, Guéant and Pu [11] provide econometric support for the win-curve assumption we make, Hendershott and Madhavan [16] document the sealed-bid mechanism itself, and Butz and Oomen [17] document electronic FX dealers skewing on flow at flat inventory, the practice that Corollary 2 derives rather than assumes.

2 Model for an imbalanced dealer market

Trade opportunities arrive as a Poisson process with mean inter-arrival time τ. Each arrival is a customer, met simultaneously by quotes from several market makers, and the trade is done at the winning quote. At each arrival there is a chance $q \in (0, 1)$ that the customer is a seller (a *buy opportunity* for our market maker) and chance $1 - q$ that the customer is a buyer. All trades are of size s. There is a fair market price commonly discerned by all dealers; in the event of a buy opportunity our market maker submits a markdown $m^{\downarrow} > 0$ below it, and in the event of a sell opportunity a markup $m^{\uparrow} > 0$ above it. She wins if her response beats the best competing response, whose displacement beyond fair value has distribution function F, survival function $1 - F$, and hazard rate $h(z)$.

We take the hazard constant, so that the best competitor is exponentially distributed with

mean

$$w = 1/h,$$

which we call the *market width*.[1] Every won trade concedes a fixed amount ϵ to adverse selection: when we mark up an item by m we assume we make only $m - \epsilon$.

Holding inventory x costs $c(x) \geq 0$ per unit time with $c(0) = 0$.[2] The game is played forever: we study the ergodic, steady state policy.

The model deliberately leaves two things out: arrivals may cluster, and the imbalance q may itself fluctuate. The symmetry is exact in the stationary core, which is what we isolate here.

Remark 1 (The fixed fair price is a frame, not a restriction)**.** *Suppose instead that the fair price F_t is a square-integrable martingale with $d\langle F\rangle_t = \sigma^2\,dt$, independent of the enquiry and fill filtration, and quotes are made relative to it; inventory x_{t-} is bounded and predictable. Cash profit then decomposes pathwise into the model's margin flow plus $M_T = \int_0^T x_{t-}\,dF_t$. The term M_T has zero conditional expectation given any state and action, so it alters no Bellman comparison and selects the same policy; $\mathbb{E}[M_T^2] = \sigma^2\,\mathbb{E}\int_0^T x_t^2\,dt = O(T)$, so $M_T/T \to 0$ in L^2 and the long-run gain is unchanged. (Boundedness of x alone would not suffice for an arbitrary martingale; the quadratic-variation rate is what controls M_T.)*

A risk-averse dealer is not indifferent to the term's variance, which by the Itô isometry is $\sigma^2\,\mathbb{E}[x^2]$ per unit time, with σ^2 the price-increment variance. Under a long-run mean–variance criterion (a variance-rate penalty with coefficient λ), the entire price risk of the moving frame is therefore the quadratic inventory charge $(\lambda\sigma^2/2)\,x^2$: a carrying cost c of exactly the canonical form, arising as an identity rather than an assumption. The equivalence requires fair-price increments independent of fills; fill-correlated price moves and state-dependent adverse selection are outside the present model, and ϵ represents only the fixed reduced-form concession assumed here. The accompanying `verify_moving_price.py` *checks the decomposition, the vanishing rate, the gain invariance for optimal and suboptimal policies, and the isometry.*

2.1 Indifference liquidation markup

Let $-\nu$ denote the average-reward relative value of inventory, normalized by $\nu(0) = 0$; only differences of ν enter the policy, and under the normalization ν may take either sign. An additively shifted version of ν has a concrete reading: a liquidation cost the market maker could pay at any time to unload her inventory, she being indifferent between paying it and carrying on — the dealer-to-dealer "haircut" at which she would clear her book. The function ν characterizes her policy, and two discrete derivatives of it will do all the work:

$$S(x) = \frac{\nu(x+s) - \nu(x-s)}{2s}, \qquad C(x) = \frac{\nu(x+s) - 2\nu(x) + \nu(x-s)}{2s},$$

the slope and convexity of the indifference cost, per unit.

[1] Remark 6 weakens the constancy to the strike range the transformation traverses. Note that w is the reciprocal cover hazard estimated from the win curve — a one-sided competitive scale, not the dealer's own quoted bid–offer width.

[2] Inventory is a real number, negative values accommodating short positions. The sign restriction on c is an economic reading; the algebraic transformation below does not require it.

2.2 Optimal markups

Define break-even markups, which we call *strikes*: the response levels at which winning the enquiry leaves the market maker indifferent,

$$K^{\downarrow}(x;s) = \epsilon + \frac{\nu(x+s) - \nu(x)}{s}, \qquad K^{\uparrow}(x;s) = \epsilon + \frac{\nu(x-s) - \nu(x)}{s},$$

for buying and selling respectively.[3] The best response solves $\sup_m (m-K)\big(1-F(m)\big)$, and evidently the optimal choice of markup lies precisely where the increase in profit exactly offsets the possibility of losing the trade. For say we increased the markup by Δm; then

$$\overbrace{\big(m^{\uparrow} - K^{\uparrow}(x;s)\big)}^{existing\ benefit} \quad \overbrace{h(m^{\uparrow})\,\Delta m}^{chance\ of\ losing\ it} \quad = \quad \overbrace{\Delta m}^{increase\ in\ profit} \quad .$$

Every interior optimizer therefore satisfies $1 = (m-K)\,h(m)$, a necessary first-order condition independent of the distributional assumption on the inside market. When the objective is unimodal — as it is under constant hazard — this condition, together with comparison to the floor at $m = 0$, characterizes the best response:

$$m^{\uparrow}(x;s) \;=\; \max\Big(\frac{1}{h\big(m^{\uparrow}(x;s)\big)} + K^{\uparrow}(x;s),\; 0\Big), \tag{1}$$

and likewise for $m^{\downarrow}$ with $K^{\downarrow}$. Under the constant hazard assumption this reads straightforwardly:

$$\text{“markup} \;=\; \text{market width} \;+\; \text{adverse selection} \;+\; \text{marginal inventory cost”}$$

though of course the coupled quantities ν and the markups remain to be found. The first order condition itself is Lerner logic with survival-function demand and is classical; the market making content lies in the indifference-adjusted strike. Away from the region where the $\max(\cdot, 0)$ binds, the two responses arrange themselves around a displaced midpoint:

$$\text{half-width}(x) = \Delta + C(x), \qquad \text{mid}(x) = \text{fair value} - S(x), \qquad \Delta = w + \epsilon. \tag{2}$$

Slope of the inventory cost is the skew; convexity of the inventory cost is the discretionary width. These identities are exact, and the second runs both ways: a dealer's quoting reveals the first two finite differences of her indifference value ν, whether or not she has ever written it down — and hence ν up to normalization, when enough inventory states are observed.[4]

2.3 The consistency equation

The market maker can remain indifferent in two ways: liquidate now and re-enter, or carry to the next opportunity. Comparing the two yields a functional equation for ν. Write $G(K)$ for the value of the next enquiry to a dealer whose strike is K: with exponential markups

$$G(K) \;=\; \sup_m\,(m-K)\,e^{-hm} \;=\; \tfrac{1}{h}\,e^{-1-hK} \quad \text{on the interior branch } K \geq -w,$$

[3] $K^{\uparrow}$ may well be negative for positive inventory: a loaded dealer will pay to get out.

[4] Recovering the primitive carrying cost c from ν requires, in addition, the consistency equation and the model's arrival and win-curve parameters.

with the floor $G(K) = -K$ below it. In average-reward form, with ρ_q the gain, the equation is

$$\rho_q \;=\; -\,c(x) \;+\; \tfrac{s}{\tau}\,H_q^{\nu}(x), \qquad H_q^{\nu}(x) \;=\; q\,G\big(K^{\downarrow}(x)\big) + (1-q)\,G\big(K^{\uparrow}(x)\big),$$

the strikes computed from ν, and an unavailable direction's term omitted from H_q^{ν} at a boundary state. Subtracting the equation's value at $x = 0$ gives the centered form used below; on the interior exponential branch it reads: for every interior state under consideration,

$$\frac{\tau\,c(x)}{s} \;=\; \frac{e^{-1-h\epsilon}}{h}\Big[e^{-hC(x)}\Big(q\,e^{-hS(x)} + (1-q)\,e^{hS(x)}\Big) \;-\; \big(\text{same expression at } x=0\big)\Big]. \quad (3)$$

In words: the direct cost of carrying x must be exactly offset by the differential option value of the next trading opportunity at x versus at zero inventory. This is why common-sense holding cost arguments, mean holding time multiplied by carrying cost and so forth, can overstate the cost of inventory and encourage overly defensive quoting. They ignore that a larger inventory makes the next enquiry *more* valuable, since the potential to get out helps more.

3 The symmetry

The paper rests on an elementary identity: for any S,

$$q\,e^{-hS} + (1-q)\,e^{hS} \;=\; 2\sqrt{q(1-q)}\,\cosh\big(h(S-\delta)\big), \qquad \delta = \frac{1}{2h}\,\log\frac{q}{1-q}. \quad (4)$$

An imbalanced mixture of exponentials is a balanced one in disguise: translated by δ, and rescaled by $2\sqrt{q(1-q)} \le 1$. Write $D(q) = 2\sqrt{q(1-q)}$ for the rescaling and $\gamma = \frac{1}{h}\log\frac{1}{D(q)} \ge 0$ for the same rescaling expressed in markup units.

Three standing assumptions scope what follows.

1. *Finite lattice.* The inventory state space is the finite lattice $\{-Ns, \dots, Ns\}$ reached from the initial inventory, with no inventory-increasing trade offered at the corresponding boundary, so the average-reward problem is a finite-state MDP.

2. *Normalization.* Relative values are normalized by $\nu(0) = 0$. The theorem is a statement about solution sets and assumes no uniqueness.[5]

3. *Interiority.* Every *available* quote is interior, $m = w + K > 0$, in both the imbalanced problem and its balanced image; the tilt makes the restriction bind earlier on one side (see Remark 2).

The equivalence is between relative values and policies, with the average reward scaling accordingly.

Theorem 1 (The imbalanced problem compresses onto the balanced one)**.** *Under the standing assumptions, (ρ_q, ν_q) solves the steady state problem with arrival imbalance q and carrying cost $c(\cdot)$ if and only if $\bar{\nu}(x) = \nu_q(x) - \delta x$ solves the perfectly balanced problem ($q = \frac{1}{2}$) with carrying cost $M(q)\,c(\cdot)$ and average reward $M(q)\,\rho_q$, where*

$$M(q) \;=\; e^{h\gamma} \;=\; \frac{1}{2\sqrt{q\,(1-q)}} \;\ge\; 1. \quad (5)$$

[5] Uniqueness of the balanced solution is invoked only where a corollary needs it (Corollaries 2 and 4); the zero-carry equation's nonidentifiability is Remark 4.

Consequently $S_q(x) = \bar{S}(x) + \delta$ *and* $C_q(x) = \bar{C}(x)$*, and for every available direction the quotes map as*

$$m_q^{\downarrow}(x) = \bar{m}^{\downarrow}(x) + \delta, \qquad m_q^{\uparrow}(x) = \bar{m}^{\uparrow}(x) - \delta :$$

at every two-sided inventory state the physical width equals the balanced-at-$M(q)c$ *width and the quote midpoint is translated by* $-\delta$*. The map* $(\rho_q, \nu_q) \mapsto (M(q)\rho_q,\ \nu_q - \delta x)$ *is thus a bijection between the normalized interior solution sets of the two problems; it assumes no uniqueness and remains valid at* $c \equiv 0$*, where* $M(q)\, c = 0$*.*

Proof. Let $\bar{\nu}(x) = \nu_q(x) - \delta x$. For every available direction the strikes translate: $K_q^{\downarrow}(x) = \bar{K}^{\downarrow}(x) + \delta$ and $K_q^{\uparrow}(x) = \bar{K}^{\uparrow}(x) - \delta$. On the interior exponential branch, using $e^{h\delta} = \sqrt{q/(1-q)}$,

$$q\, G\big(\bar{K}^{\downarrow} + \delta\big) = q\, e^{-h\delta}\, G\big(\bar{K}^{\downarrow}\big) = \sqrt{q(1-q)}\, G\big(\bar{K}^{\downarrow}\big) = \tfrac{D(q)}{2}\, G\big(\bar{K}^{\downarrow}\big),$$

$$(1-q)\, G\big(\bar{K}^{\uparrow} - \delta\big) = \tfrac{D(q)}{2}\, G\big(\bar{K}^{\uparrow}\big).$$

At a boundary state the unavailable direction's term is omitted from both Hamiltonians, so the scaling holds termwise at *every* state:

$$H_q^{\nu_q}(x) \;=\; D(q)\, H_{1/2}^{\bar{\nu}}(x).$$

Multiplying the full average-reward equation $\rho_q = -c(x) + \frac{s}{\tau} H_q^{\nu_q}(x)$ by $M(q) = 1/D(q)$ gives

$$M(q)\, \rho_q \;=\; -\, M(q)\, c(x) \;+\; \tfrac{s}{\tau}\, H_{1/2}^{\bar{\nu}}(x)$$

at every state: the balanced average-reward equation at carrying cost $M(q)c$, with gain $M(q)\rho_q$ and relative value $\bar{\nu}$. Each step is reversible, giving the equivalence, including the boundary rows. The quote map follows from the interior first-order condition $m = w + K$: the tilt adds δ to the buy-side strike and subtracts it from the sell side, and on two-sided states the shifts cancel in the width. On two-sided states the scalar manifestation of the termwise scaling is the cosh identity: substituting (4) into (3) and writing $S_\delta(x) = S(x) - \delta$ gives

$$\frac{\tau\, c(x)}{s} = \frac{e^{-1-h\epsilon-h\gamma}}{h}\Big[e^{-hC(x)} \cosh\big(hS_\delta(x)\big) - e^{-hC(0)} \cosh\big(hS_\delta(0)\big)\Big],$$

the balanced consistency equation in the pair (C, S_δ) with the cost side inflated by $e^{h\gamma} = M(q)$. □

Remark 2 (Where interiority fails)**.** *Interiority can fail at extreme or locally expensive inventory states, where the floor in* (1) *clips. In the convex cost families examined in the accompanying certificates it fails first at a one-sided boundary state: only one side trades there, so the relative value of escaping can exceed* $w + \epsilon - \delta$ *once the imbalance is strong, clipping the optimal boundary sell quote to zero at every cost level, including* $c \to 0$*. A carrying cost with a spike at an interior state can instead clip that state first. The admissible region of* (q, N, ϵ, c) *can therefore be restrictive, and the certificates assert membership rather than assume it.*

At the certificate configuration itself the assumptions are not merely asserted: an interval-Newton (Krawczyk) certificate accompanying the paper proves existence, local uniqueness, and interiority of the solution and of its balanced image, with outward-rounded arithmetic — pointwise, and by interval continuation for every $q \in [0.55, 0.605]$ *at the certificate cost.*

Corollary 1 (Imbalance is a time change of the inventory clock)**.** *Under the quote map of Theorem 1, both fill rates scale by the single factor $D(q)$:*

$$q\,e^{-hm_q^{\downarrow}} = \sqrt{q(1-q)}\;e^{-h\bar{m}^{\downarrow}}, \qquad (1-q)\,e^{-hm_q^{\uparrow}} = \sqrt{q(1-q)}\;e^{-h\bar{m}^{\uparrow}},$$

each balanced directional probability being $\frac{1}{2}$. In generator form $\mathcal{L}_{q,c} = D(q)\,\mathcal{L}_{1/2,\,M(q)c}$ under the corresponding policies: the imbalanced inventory process has the same embedded jump chain, and (subject to positive recurrence) the same stationary inventory distribution, as the balanced-at-$M(q)c$ process, with its transition clock run at rate $D(q) \le 1$. Any positive calendar-time carrying cost is therefore amplified by $M(q) = 1/D(q)$ per effective inventory transition. Imbalance cannot create carrying cost when $c \equiv 0$: $M(q)\cdot 0 = 0$.[6]

Remark 3 (The overhead frame, and what γ is)**.** *Since $D(q)\,e^{-hm} = e^{-h(m+\gamma)}$, the clock slowdown can be written algebraically as the overhead shift $\epsilon \mapsto \epsilon + \gamma$ at fixed h: γ is the logarithmic slowdown expressed in markup units. This is a reparameterization of the Bellman operator, not an additional widening of the dealer's physical quotes: if m_{over} denotes the quotes of that artificial balanced-overhead problem, the imbalanced quotes are $m_q^{\downarrow} = m_{\text{over}}^{\downarrow} + \delta - \gamma$ and $m_q^{\uparrow} = m_{\text{over}}^{\uparrow} - \delta - \gamma$, the artificial γ removed in the mapping back.*

Corollary 2 (A flat book still skews, under symmetric costs)**.** *Assume additionally $c(-x) = c(x)$, a symmetric state space and boundary, and uniqueness of the admissible balanced solution. Then the balanced solution is even, $\bar{S}(0) = 0$, hence*

$$S_q(0) \;=\; \delta \;=\; \frac{w}{2}\,\log\frac{q}{1-q} \;\approx\; 2\,w\left(q - \tfrac{1}{2}\right) \quad \textit{for mild imbalance.} \tag{6}$$

When customers are predominantly sellers the market maker marks her midpoint down by δ before holding any inventory at all. Skew is a flow phenomenon first and an inventory phenomenon second. Without cost symmetry the general formula is $S_q(0) = \bar{S}(0) + \delta$, and a dealer whose long and short inventories cost differently may skew at zero inventory even under balanced flow.

Corollary 3 (Parity without cost symmetry)**.** *For arbitrary c satisfying the standing assumptions at both q and $1-q$, since $M(q) = M(1-q)$ both problems share one balanced image, so*

$$S_q(x) - S_{1-q}(x) = 2\delta, \qquad C_q(x) = C_{1-q}(x).$$

The imbalance-induced skew component is antisymmetric around the common balanced component, without requiring cost symmetry; the total *skew's sign flip additionally needs the balanced component small.*

Corollary 4 (Skew is first order in imbalance; width is second order)**.** *Expanding about $q = \frac{1}{2}$,*

$$\delta \approx 2w\Big(q - \tfrac{1}{2}\Big), \qquad \gamma \approx 2w\Big(q - \tfrac{1}{2}\Big)^2, \qquad M(q) \approx 1 + 2\Big(q - \tfrac{1}{2}\Big)^2.$$

Provided the balanced solution branch remains interior and locally unique for cost scales between 1 and $M(q)$, and depends differentiably on the scale, a mild imbalance moves the midpoint

[6] The formal multiplier diverges as $q \to 0$ or 1, though the interior representation breaks down before the endpoints. The comparison is with the balanced dealer at cost $M(q)c$; relative to the balanced dealer at cost c the policy itself also changes, so observed throughput is not simply rescaled by $D(q)$ against that benchmark.

linearly and the width only quadratically, and the width response flows entirely through the multiplied carry: at every two-sided state, exactly,

$$W_q(x) - W_{1/2,c}(x) \;=\; 2\big[\,\bar{C}_{M(q)c}(x) - \bar{C}_c(x)\,\big],$$

the balanced convexity's comparative static in the cost scale, for which Section 4 gives a closed small-skew approximation. This accords with the practitioner's instinct to skew into flow first and adjust width only when the market becomes seriously one-sided.[7]

Remark 4 (Zero carry: the interior equations alone are nonidentifying)**.** *At $c \equiv 0$ the centered two-sided interior equations alone are nonidentifying: every linear tilt $\nu = bx$ whose quotes remain interior solves them. This does not invalidate Theorem 1, which concerns the full finite-lattice average-reward equation and maps zero-carry solutions to zero-carry solutions — $M(q) \cdot 0 = 0$, so imbalance does not manufacture carrying cost from nothing. On the finite lattice the one-sided boundary rows select a branch; in the configurations studied numerically they select the optimal tilted branch, with $S_q(0) = \delta$. On an unbounded domain an additional recurrence or transversality condition would be required instead.*

Remark 5 (The adjustment is simple; the solution is not)**.** *The theorem neither requires nor provides a closed form. In general ν must be computed numerically, and everything about its shape depends on c. What the symmetry provides is the correction: whatever the balanced solution at cost $M(q)c$ is, the imbalanced one is that same object tilted by δx. Every imbalanced instance reduces to a balanced instance: one balanced* solver *covers every level of imbalance, one solve per multiplier $M(q)$.*

Remark 6 (Exponentiality is needed only locally)**.** *The constant hazard need not hold globally. The distribution of the best competing quote enters through one object only: the value of the next enquiry as a function of its strike, $G(K) = \sup_m (m-K)\big(1-F(m)\big)$. For any smooth F, wherever the optimizer is interior, unique, and differentiable in K, the envelope theorem gives $G'(K) = -(1 - F(m^*(K)))$, and dividing by G,*

$$-\frac{d}{dK}\,\log G(K) \;=\; \frac{1}{m^*(K) - K} \;=\; h\big(m^*(K)\big).$$

Exactness requires $\log G$ to be affine over the translated strike ranges used by the visited inventory states; constant optimizer hazard throughout those ranges is sufficient, and observing the hazard at finitely many quotes is not. If the hazard varies slowly there and the quotes remain interior, $\log G$ is approximately affine, suggesting an $O(\eta)$ perturbation of the exponential symmetry with η the hazard's relative variation; a stability estimate for the consistency equation, which we do not provide here, would be required to make this exact. We checked the scaling numerically in `verify_local_exponentiality.py`*, which accompanies the paper: the error in the zero-inventory skew is roughly a quarter to a third of the hazard's relative variation across the visited strikes, and halving the perturbation multiplies the error by* 1.6–1.9*, consistent with first-order scaling.*

[7] The sign of the response follows the balanced convexity's cost-scale sensitivity: positive on the small-skew branch $hC_0 < 2$ with $C_0 > 0$, second order in general. The skew response involves no parameter beyond the observable width w; the width response involves in addition this cost-scale sensitivity.

Remark 7 (The algebra is old)**.** *We arrived at the substitution in the proof independently and noticed its provenance only afterwards. It is a birth–death symmetrization, and the geometric mean of the rates it produces is familiar from the spectral theory of birth and death processes [19, 20] and from the transient analysis of the simple queue [21]. It could equally be applied to the linear systems of the limit order book literature with asymmetric intensities, though to our knowledge nobody has done so. What we claim is the economics: after a skew translation, imbalance is a uniform slowing of the inventory clock — equivalently, a multiplication of positive carrying costs — with the constants tied to a single observable.*

4 Constant width solutions, and where CWLS sits

The slope and convexity are not independent controls: directly from their definitions,

$$S(x+s) - S(x) \;=\; C(x) + C(x+s), \tag{7}$$

so on the inventory lattice a constant convexity $C(x) \equiv C_0$ forces *affine* skew, $S(x) = \frac{2C_0}{s}x + b$: no single indifference cost generates a constant width with nonlinear skew. With that understood, the consistency equation can still be inverted pointwise,

$$S_\delta(x) \;=\; \frac{1}{h}\cosh^{-1}\Big(e^{hC_0}\,\Omega\,c(x) + \cosh\big(hS_\delta(0)\big)\Big), \qquad \Omega = \tfrac{\tau h}{s}\,e^{1+h\epsilon+h\gamma}, \tag{8}$$

a pointwise inversion whose right side is in general *nonlinear* in $c(x)$ — implementable as a reduced-form heuristic, but by (7) a genuine value-function solution only where it is affine, which is what pins the cosh cost family below. The branch is chosen so that S_δ is increasing; for even c, on the centered odd branch $S_\delta(0) = 0$, $S_\delta(x) = \operatorname{sgn}(x)\,\frac{1}{h}\operatorname{arcosh}(\cdot)$. There, for small argument the right side is approximately proportional to $\sqrt{c(x)}$, so a linear component in c (a physical holding fee) puts a square-root kink in the skew at the origin, and funding costs or equity hurdles bend it elsewhere.

Where, then, is CWLS exactly optimal? Requiring $C(x) \equiv C_0$ together with consistency of (3) and an even carrying cost forces

$$\nu(x) \;=\; \frac{C_0}{s}\,x^2 \;+\; \delta\,x \;+\; C_2, \tag{9}$$

with the imbalance appearing as the *linear term of the indifference cost*, and feeding (9) back through (3) pins the cost of carry to the shape $c(x) \propto \cosh(2hC_0x/s) - 1$. The quadratic term prices inventory risk; the linear term prices the flow; and the skew $S(x) = 2\frac{C_0}{s}\,x + \delta$ is linear with an intercept.

The corner also prices the width response to imbalance. Exactly, by Theorem 1, $W_q(x) - W_{1/2,c}(x) = 2[\bar{C}_{M(q)c}(x) - \bar{C}_c(x)]$: the balanced convexity's comparative static in the cost scale, evaluated at scale $M(q)$. In the small-skew quadratic-cost regime the sensitivity has a closed form: the small-x expansion of the quadratic family gives $e^{-hC_0}C_0^2$ proportional to the cost scale, where $C_0 = \bar{C}_c(0)$ is the balanced baseline, hence $dC_0/d\log(\text{scale}) = C_0/(2 - hC_0)$ on the branch $hC_0 < 2$, and

$$\Delta C(0) \;\approx\; \gamma\,\frac{hC_0}{2 - hC_0}, \tag{10}$$

the approximation carrying the $O((q-\frac{1}{2})^4)$ error of linearizing the balanced convexity in $\log M$ (the relation $\log M = h\gamma$ itself is exact), together with the small-skew approximation error.

The even, sign-definite γ-shape in the imbalance survives, scaled by roughly half the ratio of discretionary to market width when $hC_0 \ll 1$; in full-width terms $\Delta W(0) = 2\Delta C(0) \approx 2C_0(q-\frac{1}{2})^2$. The accompanying scripts confirm the formula within roughly one to two percent in the tested configurations across $q \in [0.55, 0.7]$, and confirm that the imbalanced width equals the balanced-at-$M(q)c$ width to machine precision.

The interior Bellman equations therefore admit exact CWLS in one corner — balanced flow and a cosh-shaped cost of carry — a shape that is quadratic up to a relative error of $(hS_\delta(x))^2/12$; on the finite capped lattice the one-sided boundaries generate a boundary layer, so the global optimum departs from CWLS near the caps. Since the compatible cosh cost differs from its quadratic approximation by relative order $(hS_\delta(x))^2/12$, for a genuinely quadratic cost CWLS is expected to be a second-order approximation on a stable interior solution branch, which is what the accompanying script measures. On an unbounded lattice the linear skew eventually drives one quote into the floor of (1), so the corner's exactness, like the theorem's, lives on the interior region. The symmetry supplies the benchmark's missing column: under imbalance, keep the linear response to inventory computed at cost $M(q)c$, and add the intercept δ. The moral for anyone outside the corner is procedural: solve for $\nu(x)$ rather than pretend $\nu \propto x^2$ works, since the quadratic ansatz is internally inconsistent for any other cost of carry.

5 Uses

The equivalence was stated as a fact about optimal policy, but its practical content is broader, and we note five uses.

Fill ratios. At the optimum, profitability factors cleanly. By (1) the optimal response sits the distance w beyond its strike, so the *continuation-adjusted surplus* on a won trade — the one-step Bellman value, not the cash margin $s(m-\epsilon)$, which varies with inventory through the continuation term — is $s\,w$ at every level of inventory, and all variation in the one-step Bellman value per enquiry is variation in the fill ratio. Under the constant hazard the fill ratio on a quote at markup m is $e^{-m/w}$, log-linear in the markup with slope $-1/w$, which is how w is estimated from a quote archive; at the optimal response to strike K it equals $e^{-1-K/w}$. In particular a dealer facing zero adverse selection and zero marginal inventory cost should, if she quotes optimally, win $e^{-1} \approx 37\%$ of her enquiries, and persistent departures from that ratio measure her effective strike.

Testing the symmetry. Request-for-quote archives record direction, quotes, and fills. Direction counts estimate q, the win curve estimates w, with Remark 6 indicating that the local estimate is the appropriate one, and the theorem then predicts the zero-inventory skew $\delta = \frac{w}{2}\log\frac{q}{1-q}$ with no free parameter (under symmetric costs), and the width response through the carry channel: within an instrument, $[W_q(0) - W_{1/2}(0)]/C_0 \approx 2(q-\frac{1}{2})^2$ on the small-hC_0 branch of (10), where C_0 is extracted from the quoted width net of w and ϵ rather than read directly. Cross-sectionally, mids should shift at first order in the imbalance and widths at second order, the width response proportional to the discretionary width rather than the market width; cross-instrument comparisons are confounded by carrying-cost differences and are weaker tests than the within-instrument shape.

Reading a dealer's book. The identities (2) run both ways: skew and discretionary width are the first two differences of the indifference cost, so a counterparty's quoting reveals her inventory cost function, and a shift in her zero-inventory skew reveals her view of the flow.

Increasing the statistical power of learned policies. Reinforcement learning approaches to market making [12] generalize the present model and need not assume an exponential win curve. The symmetry helps them anyway, reinterpreted as an approximate heuristic: experience gathered at one imbalance transfers to every other through the correction $(\delta, M(q))$ — the skew tilt and the cost rescaling — so a learner can pool episodes across flow regimes instead of learning each regime separately. An invariance of this sort reduces the effective dimension of the policy space and can improve sample efficiency by allowing episodes from different flow regimes to be pooled after transformation.

Assessing decisions after the fact. Illiquidity implies short time series, and short time series make back-testing unreliable; the model instead prices individual decisions. Comparing a trader's response m against the model's m^* on a single enquiry:

	model trades	model misses
trader trades	$s\,(m - m^*)$	$s\,(m - m^*) + s\,w$
trader misses	$-\,s\,w$	0

Every cell is computable when the cover price (the best competing quote) is observed, as on platforms that disclose it to the winner; without a cover, the two disagreement cells are counterfactual. On interior quotes the running total is a one-step deviation score: it compares the trader's quote with the model quote at the model's strike, with the model policy resumed afterward, and is therefore exact for single deviations rather than the total advantage of a permanently different policy. It separates flow-reading (the δ term) from inventory management (the S_δ term). Profit and loss, by contrast, mixes both with luck. The constant margin under exponential demand is itself a classical property; the diagnostic built on it is not.

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